Ashiqur Rahman Rahul

AI Systems Engineer | Embedded Systems | GPU & Edge Computing

Tessiner Weg 18, 13407 Berlin | +49 1638723454 | Email | GitHub | Portfolio | LinkedIn

PROFESSIONAL SUMMARY

AI Systems Engineer with experience in AI product development, machine learning, embedded systems, GPU computing, and infrastructure-aware optimization. Currently working with the Cambridge Centre for Alternative Finance, University of Cambridge, on AI energy profiling, benchmarking, optimization strategy, and infrastructure-focused research. Interested in AI model development, AI systems engineering, embedded systems, machine learning, edge AI, intelligent infrastructure, and scalable deployment optimization.

PROFESSIONAL EXPERIENCE

AI Research Analyst - *Cambridge Centre for Alternative Finance (CCAF), University of Cambridge*

Remote, Berlin **|** 07/2025 - Present

- Developed **AI benchmarking and analytical frameworks** evaluating **6+ infrastructure and deployment metrics** including deployment & inference efficiency, reliability, infrastructure utilization, deployment latency, and energy-aware optimization across distributed AI workloads.
- Architected and developed **4+ AI-driven analytical and optimization frameworks** focused on infrastructure-aware model evaluation, deployment intelligence, benchmarking automation, and sustainable AI systems analysis for large-scale fintech research environments.
- Built **3 Python-based AI analytics pipelines**, operational dashboards, and infrastructure-aware optimization workflows using **Python, Streamlit, FastAPI, React and cloud-based tooling** to support scalable AI operations.
- Contributed to research initiatives on **sustainable AI systems**, **AI energy efficiency**, **sovereign AI infrastructure**, and operational AI economics through technical analysis, deployment evaluation, and white paper development.
- Authored and contributed to **2 white papers**, **1 technical survey**, **2 technical reports**, and **4+ research initiatives** focused on sustainable AI model development, scalable AI systems, deployment optimization, and infrastructure-aware AI analytics.

Research Intern - *Cambridge Centre for Alternative Finance (CCAF), University of Cambridge*

Remote, Berlin | 07/2024 - 06/2025

- Supported research on **AI framework development**, **AI energy efficiency**, model optimization, and infrastructure-aware AI deployment analysis across sustainable AI research initiatives.
- Assisted in developing **2 AI adoption surveys**, sustainable model optimization workflows, fine-tuning experiments, performance analysis pipelines, software environment setup, and technical research documentation.
- Contributed to **benchmarking experiments** and technical documentation, helping improve reproducibility and reduce evaluation inconsistencies across research pipelines by ~**15%**.

Research Assistant - *Hamburg University of Technology (TUHH)*

Hamburg, Germany | 04/2024 - 09/2024

- Developed edge AI and hyperspectral anomaly detection pipelines **using deep learning and transfer learning**, **improving** training accuracy by **~53%**, inference accuracy by **~28%**, and energy efficiency by **~46%** across experimental benchmarks.
- Optimized deployment and benchmarking workflows on **NVIDIA Jetson Nano, TX2, and AGX Xavier**, reducing inference latency by ~30% and improving GPU utilization efficiency.
- Conducted embedded AI performance analysis across latency, energy consumption, and deployment efficiency using **Python, CUDA, MATLAB, Linux, and C++**.

Research Assistant - *Hanbat National University*

Remote, Daejeon, South Korea **|** 06/2019 - 08/2021

- Researched **wireless communication systems**, cognitive networks, and IoT architecture across **10+ simulated scenarios**.
- Built simulation and analytics pipelines using **MATLAB, Python and C++**, reducing evaluation time by **35%** and improving energy efficiency by **48%**.
- Supported simulation, documentation, and data analysis using **MATLAB, Python, LaTeX, and Excel** for network modeling and optimization.
- Co-authored **3 peer-reviewed publications** in **IEEE, Springer, and Wiley** venues.

Journal Reviewer - *Science Publishing Group*

Remote, Rockefeller Plaza, New York, | NY 10020 09/2020 - 09/2021

- Reviewed 12+ journal articles and 3 book chapters for Advances in Wireless Communications and Networks (AWCN).

EDUCATION

M.Sc. Microelectronics and Microsystems - *Hamburg University of Technology (TUHH)*

Hamburg, Germany | 10/2021 - 04/2026

- **Specialization:** Embedded Systems and Microsystem Technology.
- **Relevant focus:** GPU architecture, energy efficiency in embedded systems, communication networks, and sustainable technology management.
- **Thesis:** *Hyperspectral Anomaly Detection on Edge Platforms: Transfer-Learning Enhanced Autoencoders with Comparative Multi-Metric Analysis*.

B.Sc. Electrical and Electronic Engineering - *BRAC University*

Dhaka, Bangladesh **|** 12/2014 - 05/2019

- **Specialization:** Communication Engineering and Computer Engineering.
- **Thesis:** *Performance Analysis of Cognitive Unmanned Aerial Vehicle*. - Link (Utilizing signal modelling, spectral efficiency analysis, and system simulation)

EXPERTISE

AI Systems Engineering | AI Infrastructure & Optimization | Embedded Systems | GPU Computing & CUDA | Edge AI | Machine Learning & Computer Vision | LLM & RAG Systems | Applied Machine Learning & Deep Learning | Energy-Efficient Computing | AI Benchmarking & Analytics | IoT & Hardware Integration | Technical Research

TECHNICAL SKILLS

Programming: Python (Advanced), MATLAB (Advanced), C/C++ (Advanced), R, SQL, Excel VBA

AI / ML / Data: Machine Learning, Deep Learning, Transfer Learning, DevOps, Model Fine-Tuning, Prompt Engineering, Computer Vision, Telemetry Analysis, Containerization (Docker), NLP, LLM Design, RAG, Data Analysis, Node.js, React, Model Optimization, ONNX Runtime, Inference Optimization, Pareto Optimization, MLOps Concepts (Advanced), AI Infrastructure Analysis, Performance Benchmarking

Embedded / Hardware: Embedded Systems, Embedded software design, IoT System Design, Hardware Setup and Simulation, ESP32 (Advanced), Raspberry Pi (Advanced), Arduino UNO (Advanced)

GPU / Edge Systems: NVIDIA CUDA (Intermediate), GPU Applications, GPU Architecture, NVIDIA Jetson Nano, Jetson TX2, AGX Xavier

Tools / Platforms: Linux (Advanced), Windows, GitHub, Visual Studio, Docker (Advanced), FastAPI (Advanced), Streamlit (Advanced), ONNX Runtime, REST APIs, PostgreSQL, GitHub Actions (Advanced), MATLAB Simulink (Intermediate), Proteus, KiCad, COMSOL, Autodesk Fusion 360.

Documentation / Research: Scientific Writing, Technical Writing, Academic Publication, Digital Research, Research Methodology, LaTeX Typesetting, Proprietary Data Handling

Office / Productivity: MS Word, Excel, PowerPoint, Access, Publisher, Visio – (All Advanced)

SELECTED PROJECTS (GitHub)

- *AI Inference Optimization & Deployment Decision Engine:* Built a multi-objective AI deployment optimization platform using Streamlit, ONNX Runtime, Pareto optimization, autoscaling simulation, and carbon-aware analytics. Link
- *AI Energy Efficiency Index:* Developed a framework for benchmarking AI inference infrastructure efficiency, energy consumption, and deployment optimization. Link
- *Signal Processing & Intelligent Analytics System:* Built a modular signal analysis and visualization platform with analytical dashboards, intelligent monitoring workflows, and data-driven evaluation pipelines. Link
- *Edge AI Hyperspectral Anomaly Detection:* Designed & developed transfer-learning autoencoder pipelines on NVIDIA Jetson devices with energy-aware benchmarking and optimization analysis.

PUBLICATIONS

- *Cognitive Unmanned Aerial Vehicle-aided Human Bond Communication System: Modeling and Performance Analysis*. IEEE Access, 2022 - Link
- *An optimization based approach to enhance the throughput and energy efficiency for cognitive unmanned aerial vehicle networks*. Wireless Networks, Springer, 2021 - Link
- *Cognitive IoT-Based Health Monitoring Scheme Using Non-Orthogonal Multiple Access*. Wiley Book Chapter, 2022 - Link

LANGUAGES & AWARDS

Languages: Bengali (Native), English (C2- Native/Fluent), German (A2- Intermediate).

Awards: Vice Chancellor's List Award, BRAC University; Dean's List Award, BRAC University.

Rahul

Berlin, Germany